

1 Analysis of the given model

The given model was trained using the following hyperparameters: $epochs = 10$, $batch\ size = 128$, $learning\ rate = 0.1$ and $optimizer = Adam$. The loss figure clearly shows two issues: the first one is an unstable training, this is suggested by the extremely steep drop in the training loss line, this problem could be solved by decreasing the learning rate value. The second problem is a bad generalization, which is shown by the instability of the validation curve.

2 Optimization

The first optimization done is reducing the learning rate at 0.001 and increasing the epochs to observe the behaviour of the model.

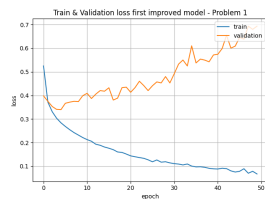


Figure 1: Loss of the first optimized model

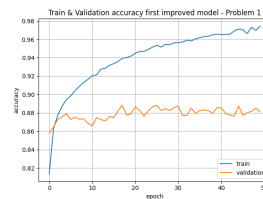


Figure 2: Accuracy of the first optimized model

Figures 1 and 2 indicate that the model is overfitting, which is a sign of poor generalization. Based on this analysis, regularization techniques were applied to the model to reduce overfitting and improve its performance.

The weights of the given model were plotted to help choose an appropriate regularization technique. It is clear that the weights of each network layer have a mean close to 0. In this case, the application of L1 regularization would be not useful because it tries to shrink weights to 0 (promoting sparsity), but the majority of the weights are already sparse because they are close to zero. Instead, L2 regularization was applied to penalize severely large weights and promote a less sparse weight distribution. This helps the model to improve generalization and reduces the risk of overfitting. After applying L2 regularization, a Dropout layer with a rate of 0.25 was added between the input layer and the first hidden layer. This was done to improve the model's generalization and reduce overfitting. At the end was applied an Early Stopping callback to stop the model when loss stop decreasing for 5 epochs by a certain delta. This value was fixed to 0.02 for two reasons: the first one is to prevent overfitting, the second one is that with $\delta = 0$ the callback did not stop the training even though the loss curves had already stabilized.

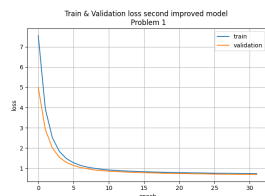


Figure 3: Loss of the second optimized model

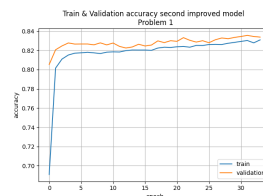


Figure 4: Accuracy of the second optimized model

Figures 3 and 4 show the final results obtained by the model. Compared to the initial model, there are visible improvements leading to better generalization and reduced overfitting.